

ENVIRONMENTAL SCIENCE

Unit 3 Comprehensive Notes

Other Environmental Pollution & Waste Management

Soil Pollution • Noise Pollution • Nuclear & Radiation Hazards

Marine Pollution • Solid Waste Management • Bioremediation • E-Waste

■ Syllabus Coverage (Unit 3)

- Soil Pollution: Soil around us, Soil water characteristics, Soil pollution
- Causes, effects & control: Noise pollution, Nuclear & Radiation hazards
- Marine Pollution: Oil spills & Ocean Acidification
- Solid Waste Management: Composting, Vermiculture, Landfills, Hazardous waste treatment
- Bioremediation technologies, Conventional techniques (land farming, constructed wetlands)
- Phytoremediation, Degradation of xenobiotics in environment
- Petroleum hydrocarbons, Pesticides, Heavy metals
- E-Wastes: Introduction, types, environmental impact, recycling, e-waste management rules

■ How to Use These Notes

- Each topic begins with a clear **definition** followed by detailed explanation.
- **Causes** → **Effects** → **Control** structure used throughout for easy revision.
- Tables used only for **Difference Between** questions.
- End of document contains **7-mark Questions & Answers** for exam preparation.
- Key Points boxes highlight must-remember facts for quick revision.

PART A: SOIL POLLUTION

1.1 Soil Around Us

Soil is the thin, uppermost layer of Earth's crust that supports life. It is a complex mixture of minerals, organic matter (humus), water, air, and living organisms. Soil is often called the 'skin of the Earth' because it acts as a bridge between the living (biotic) and non-living (abiotic) worlds.

Composition of Soil

- **Minerals (45%):** Derived from weathering of parent rock — sand, silt, clay particles.
- **Organic Matter/Humus (5%):** Decomposed plant and animal remains; provides nutrients.
- **Water (25%):** Held in pore spaces; essential for nutrient transport to plants.
- **Air (25%):** Fills pore spaces; oxygen essential for soil organisms and root respiration.
- **Living Organisms:** Bacteria, fungi, earthworms, nematodes — decomposers that recycle nutrients.

Soil Profile (Horizons)

- **O-Horizon:** Organic layer — fresh/decomposing leaf litter at surface.
- **A-Horizon (Topsoil):** Dark, fertile zone rich in humus and microorganisms.
- **B-Horizon (Subsoil):** Accumulation of leached minerals from above (illuviation).
- **C-Horizon:** Partially weathered parent rock (regolith).
- **R-Horizon:** Solid, unweathered bedrock.

1.2 Soil Water Characteristics

Soil water refers to the water present in the soil pore spaces. It is vital for plant growth, microbial activity, and the transport of dissolved nutrients.

Types of Soil Water

- **Gravitational Water:** Water that drains freely under gravity through macro-pores. Not available to plants for long.
- **Capillary Water:** Water held in micro-pores by surface tension (capillary forces). This is the *most available* form for plants.
- **Hygroscopic Water:** Thin film of water tightly bound to soil particles by adsorption; unavailable to plants.
- **Chemically Combined Water:** Water forming part of mineral crystal structure; not available to plants.

Key Soil Water Terms

- **Field Capacity:** Maximum water soil holds after excess drainage — optimum for plant growth.
- **Wilting Point:** Minimum water content below which plants wilt and cannot recover.
- **Available Water:** Water between field capacity and wilting point — the water usable by plants.
- **Soil Moisture Tension:** Force with which water is held by soil particles.

◆ KEY FACT

The relationship between soil texture and water-holding capacity is critical: Clay soils hold more water (high surface area) but drain poorly; Sandy soils drain fast but hold little water; Loam soils (mixture) are ideal for agriculture.

1.3 Soil Pollution

Soil pollution (also called land pollution or soil contamination) is the introduction of harmful substances into the soil that degrade its quality, reduce fertility, and make it unfit for plant growth, animal life, or human use.

Causes of Soil Pollution

- 1. Industrial Effluents & Solid Wastes:** Heavy metals (lead, mercury, cadmium, arsenic) from mining, smelting, electroplating. Industrial sludge dumped on land contaminates soil with toxic chemicals.
- 2. Agricultural Chemicals:** Excessive use of synthetic fertilizers (nitrates, phosphates), pesticides, herbicides, and fungicides. These persist in soil, killing beneficial microorganisms and accumulating in food chains (biomagnification).
- 3. Urban & Municipal Waste:** Improper disposal of household garbage, sewage sludge, and construction debris. Plastics are particularly dangerous — they take 100–500 years to degrade.
- 4. Radioactive Waste:** Fallout from nuclear tests, leaks from nuclear power plants, and improper disposal of radioactive medical/industrial waste contaminate soil for centuries.
- 5. Acid Rain:** SO_2 and NO_x from burning fossil fuels dissolve in rainwater forming H_2SO_4 and HNO_3 , which acidify soil, leach essential nutrients (Ca, Mg), and mobilize toxic metals.
- 6. Oil Spills & Petroleum Products:** Leaking underground storage tanks and oil pipeline accidents contaminate large areas of soil with hydrocarbons.
- 7. Deforestation:** Removal of tree cover exposes soil to erosion by wind and water, reducing topsoil depth and fertility.

Effects of Soil Pollution

- **Loss of Soil Fertility:** Toxic chemicals kill beneficial microbes and earthworms, disrupting nutrient cycles.
- **Bioaccumulation:** Heavy metals and pesticides enter food chains — concentrating in top predators (including humans).
- **Groundwater Contamination:** Pollutants leach into groundwater, polluting drinking water sources.
- **Loss of Biodiversity:** Contaminated soil cannot support normal plant communities, reducing habitat diversity.
- **Health Effects:** Exposure to contaminated soil causes cancer, kidney/liver damage, neurological disorders, and developmental problems in children.
- **Reduced Crop Yield:** Degraded soil produces lower quality and quantity of agricultural produce.
- **Desertification:** Severe soil degradation leads to conversion of fertile land into desert.

Control of Soil Pollution

- **Reduce Chemical Use:** Integrated Pest Management (IPM), organic farming, biofertilizers instead of synthetic ones.
- **Proper Waste Disposal:** Controlled landfills, composting of organic waste, recycling of materials.
- **Bioremediation:** Using microorganisms/plants to break down or immobilize pollutants.
- **Phytoremediation:** Using hyperaccumulator plants (e.g., sunflower for uranium, Indian mustard for lead) to extract heavy metals.
- **Soil Washing:** Flushing contaminated soil with water/chemical solutions to remove pollutants.
- **Legislation:** Strict enforcement of Environmental Protection Act, hazardous waste management rules.
- **Afforestation:** Planting trees to prevent soil erosion and restore fertility.

PART B: NOISE POLLUTION

2.1 Definition and Measurement

Noise pollution is defined as unwanted, unpleasant, or excessive sound that disrupts the environment and causes harmful effects on humans, animals, and the ecosystem. Sound is measured in **decibels (dB)**.

Sound Level Reference Chart (Decibel Scale)

• 0 dB — Threshold of human hearing
• 30 dB — Quiet library, whisper
• 60 dB — Normal conversation
• 85 dB — Heavy traffic (threshold of hearing damage with prolonged exposure)
• 100 dB — Construction site, jackhammer
• 120 dB — Thunderclap, rock concert (pain threshold)
• 140 dB — Jet engine at takeoff (immediate danger)

2.2 Causes of Noise Pollution

- 1. Transportation:** Road traffic (honking vehicles), railways, aircraft — the largest source in urban areas.
- 2. Industrial Sources:** Factories, machinery (compressors, generators, heavy equipment) operating near residential areas.
- 3. Construction Activities:** Pile driving, drilling, demolition, and use of heavy machinery.
- 4. Social/Cultural Events:** Loudspeakers, festivals (firecrackers), religious gatherings, public address systems.
- 5. Household Sources:** TV, music systems, mixers, air conditioners, refrigerators.
- 6. Agricultural Activities:** Tractors, threshers, irrigation pump sets.

2.3 Effects of Noise Pollution

Effects on Humans

- **Auditory Effects:** Tinnitus (ringing in ears), Noise-Induced Hearing Loss (NIHL), temporary/permanent deafness.
- **Cardiovascular Effects:** Elevated blood pressure, increased heart rate, risk of heart disease.
- **Psychological Effects:** Stress, anxiety, irritability, depression, sleep disturbances.
- **Cognitive Effects:** Reduced concentration, impaired learning in children, reduced productivity.
- **Physiological Effects:** Headaches, fatigue, hormonal imbalances (increased adrenaline and cortisol).

Effects on Animals and Wildlife

- Marine animals (whales, dolphins) lose their sonar navigation — causing beaching events.
- Birds abandon nesting sites near noisy areas, affecting reproduction.

- Noise disturbs feeding, mating, and communication in wildlife.

2.4 Control of Noise Pollution

- **Source Control:** Silencers on vehicles/machines, anti-vibration mountings, use of quiet equipment.
- **Path Control:** Sound barriers (walls, embankments), sound-absorbing materials, green belts (trees absorb noise).
- **Receiver Control:** Ear protection (earmuffs, earplugs), soundproofing of buildings.
- **Zoning:** Industrial zones separated from residential areas; no honking zones near hospitals and schools.
- **Legislation:** Noise Pollution (Regulation and Control) Rules, 2000 — sets permissible limits for different zones.
- **Public Awareness:** Education about noise hazards; banning unnecessary horn use.

♦ IMPORTANT FOR EXAM

Permissible Noise Levels (India, Noise Rules 2000): Industrial area: Day 75 dB / Night 70 dB; Commercial area: Day 65 dB / Night 55 dB; Residential area: Day 55 dB / Night 45 dB; Silence zone (hospitals/schools): Day 50 dB / Night 40 dB.

PART C: NUCLEAR & RADIATION HAZARDS

3.1 Introduction to Radiation

Radiation is energy emitted as electromagnetic waves or particles. **Ionizing radiation** has enough energy to remove electrons from atoms, causing chemical changes in living tissues. Nuclear and radiation hazards arise from radioactive substances and nuclear processes.

Types of Ionizing Radiation

- **Alpha (α) radiation:** Helium nuclei; low penetration (stopped by paper/skin) but very harmful if inhaled/ingested.
- **Beta (β) radiation:** High-speed electrons; penetrate skin (stopped by plastic/aluminum); hazardous if internal.
- **Gamma (γ) radiation:** High-energy electromagnetic radiation; highly penetrating (requires lead/concrete shielding); most dangerous external hazard.
- **X-rays:** Similar to gamma but lower energy; medical uses pose minimal risk but require precautions.
- **Neutron radiation:** Released during nuclear fission; penetrating and induces radioactivity in materials.

3.2 Sources of Nuclear & Radiation Pollution

1. **Nuclear Power Plants:** Routine operations release small amounts of radioactive gases/water; accidents (Chernobyl 1986, Fukushima 2011) release massive radioactivity.
2. **Nuclear Weapons Testing:** Atmospheric nuclear tests (1945–1980s) spread radioactive fallout (Strontium-90, Cesium-137) globally.
3. **Medical Uses:** X-ray machines, CT scanners, radiotherapy equipment, and radioisotopes used in diagnosis contribute to occupational and patient exposure.
4. **Industrial Uses:** Radioactive sources used in gauges, sterilization, and non-destructive testing can cause exposure if mishandled.
5. **Mining of Radioactive Ores:** Uranium and thorium mining releases radon gas and radioactive dust.
6. **Cosmic Radiation:** High-energy particles from space; increases with altitude (significant for frequent flyers and airline crew).
7. **Natural Background Radiation:** From radioactive rocks (radon from granite), soil, and food (potassium-40 in bananas).

3.3 Effects of Nuclear/Radiation Pollution

Somatic Effects (Effects on Exposed Individual)

- **Acute Radiation Syndrome (ARS):** High doses (>1 Gy) cause nausea, vomiting, hair loss, bone marrow suppression, and death.
- **Cancer:** Radiation is a potent carcinogen — leukemia, thyroid cancer, lung cancer are most common radiation-induced cancers.
- **Cataracts:** Lens of the eye is particularly sensitive; high doses cause radiation cataracts.
- **Skin Burns:** Radiation burns (erythema) at exposure sites.

Genetic Effects (Effects on Future Generations)

- DNA mutations in reproductive cells can be passed to offspring.
- Chromosomal aberrations — increased rate of birth defects, stillbirths, and hereditary diseases.
- Radiation can cause point mutations altering gene function.

Environmental Effects

- Radioactive contamination of soil, water, and air affects entire ecosystems.
- Bioaccumulation of radioactive isotopes (e.g., Strontium-90 in bones, Cesium-137 in muscle tissue) through food chains.
- Long-term contamination of large land areas (Chernobyl exclusion zone, 2,600 km²).

3.4 Control of Nuclear/Radiation Hazards

- **Distance:** Radiation intensity decreases with square of distance — maintain safe distances from sources.
- **Shielding:** Use lead aprons, lead-lined walls, concrete barriers based on radiation type.
- **Time Limitation:** Minimize duration of exposure to radiation sources.
- **Personal Protective Equipment:** Lead aprons, dosimeters (to monitor exposure), protective clothing.
- **Safe Disposal of Radioactive Waste:** Low-level waste — near-surface disposal; High-level waste — deep geological repositories.
- **International Treaties:** Nuclear Non-Proliferation Treaty (NPT); Comprehensive Test Ban Treaty (CTBT).
- **Emergency Preparedness:** Evacuation plans, iodine tablet distribution to prevent thyroid radiation absorption.

PART D: MARINE POLLUTION — OIL SPILLS & OCEAN ACIDIFICATION

4.1 Marine Pollution — Overview

Marine pollution is the introduction of harmful substances or energy into the ocean from human activities, causing adverse effects on marine ecosystems, biodiversity, and human health.

4.2 Oil Spills

An oil spill is the release of liquid petroleum hydrocarbons into the ocean or other water bodies due to human activity. It is one of the most visible and damaging forms of marine pollution.

Sources of Oil Pollution in Oceans

- **Tanker Accidents:** Collision or grounding of oil tankers — e.g., Exxon Valdez (1989), Deepwater Horizon (2010).
- **Offshore Drilling:** Blowouts and leaks from oil platforms.
- **Routine Shipping:** Bilge pumping, tank cleaning operations.
- **Pipeline Leaks:** Corrosion or damage to undersea/coastal pipelines.
- **Natural Seeps:** Natural discharge of oil from seafloor seeps (~600,000 tons/year).
- **Runoff:** Oil-contaminated stormwater runoff from roads, vehicles entering rivers and coastal waters.

Effects of Oil Spills

- **Seabirds:** Oil mats feathers, destroying insulating and waterproofing properties → hypothermia, drowning. Birds ingest oil while preening → liver and kidney damage.
- **Marine Mammals:** Seals, sea otters — fur loses insulation; whales inhale vapors causing lung damage.
- **Fish & Marine Life:** Toxic compounds (PAHs) kill fish larvae and eggs; reduces oxygen in water.
- **Coral Reefs:** Smothered by oil, causing bleaching and death of coral polyps.
- **Mangroves & Coastal Ecosystems:** Oil clogs mangrove root pores, killing trees; destroys mudflat habitats.
- **Fisheries & Livelihoods:** Massive economic losses to fishing communities; contaminated seafood is unsafe to consume.
- **Tourism:** Oiled beaches reduce tourism revenue for coastal communities.

Cleanup Methods for Oil Spills

- **Booms:** Floating barriers to contain the oil slick and prevent spreading.
- **Skimmers:** Mechanical devices that skim oil off the water surface.
- **Sorbents:** Materials (straw, peat, synthetic foams) that absorb oil.
- **Chemical Dispersants:** Chemicals (e.g., Corexit) that break oil into smaller droplets — controversial as they are also toxic.
- **In-situ Burning:** Controlled burning of surface oil slicks — removes oil quickly but creates air pollution.
- **Bioremediation:** Adding oil-degrading bacteria (*Pseudomonas*, *Alcanivorax*) or nutrients to stimulate natural bacterial breakdown.
- **Manual Cleanup:** Volunteers cleaning oiled shorelines with shovels and absorbent materials.

4.3 Ocean Acidification

Ocean acidification is the ongoing decrease in the pH of Earth's oceans caused by the absorption of carbon dioxide (CO₂) from the atmosphere. Since the Industrial Revolution, ocean pH has dropped from 8.2 to approximately 8.1 — a 26% increase in acidity.

Chemistry of Ocean Acidification

When CO₂ dissolves in seawater, it forms carbonic acid (H₂CO₃), which dissociates to release hydrogen ions (H⁺), reducing pH. This also reduces the concentration of carbonate ions (CO₃²⁻) essential for shell formation:



Effects of Ocean Acidification

- **Shell Formation:** Oysters, clams, sea urchins, and pteropods cannot form/maintain shells in acidic water — shells dissolve.
- **Coral Bleaching & Death:** Corals cannot build calcium carbonate skeletons; reef growth rates decrease; existing reefs dissolve.
- **Food Web Disruption:** Loss of pteropods (tiny sea snails) — a key food source for salmon, herring, and whales.
- **Fish Behavior:** Impairs sense of smell (used to detect predators/prey) and otolith (ear bone) development in fish.
- **Biodiversity Loss:** Projected loss of 50–70% of coral reefs if CO₂ levels continue rising.

Solutions to Ocean Acidification

- **Reduce CO₂ Emissions:** Transition to renewable energy; energy efficiency; carbon capture technologies.
- **Ocean Alkalinity Enhancement:** Adding alkaline minerals (limestone/olivine) to seawater to neutralize acidity — experimental.
- **Marine Protected Areas:** Reducing other stressors (pollution, overfishing) gives coral reefs better resilience.
- **Seaweed Farming:** Macroalgae absorbs CO₂ — large-scale seaweed farming can locally reduce acidity.

PART E: SOLID WASTE MANAGEMENT

5.1 Introduction to Solid Waste

Solid waste refers to any non-liquid, non-gaseous material that is discarded by society as unwanted. With urbanization and industrialization, solid waste generation has increased dramatically. India generates approximately 62 million tonnes of municipal solid waste annually.

Categories of Solid Waste

- **Municipal Solid Waste (MSW):** Household garbage, food waste, paper, plastic, glass.
- **Industrial Waste:** Manufacturing by-products, slags, chemicals.
- **Hazardous Waste:** Toxic, flammable, reactive, or corrosive materials.
- **Biomedical Waste:** Hospital waste — needles, blood-soaked materials, expired drugs.
- **Agricultural Waste:** Crop residues, animal manure.
- **E-Waste:** Discarded electronic equipment (covered in detail in Part G).

5.2 Composting

Composting is the controlled biological decomposition of organic matter (food scraps, yard waste) by microorganisms (bacteria, fungi, actinomycetes) under aerobic conditions to produce compost — a stable, humus-like material used as soil amendment.

Types of Composting

- **Open Windrow Composting:** Organic waste is piled in long rows (windrows) and turned periodically for aeration. Suitable for large-scale municipal composting.
- **Aerated Static Pile Composting:** Organic waste mixed with bulking agent (wood chips); air blown through the pile via pipes. Faster process.
- **In-Vessel Composting:** Material placed in enclosed container with controlled temperature, moisture, and aeration. Fastest method; suitable for hospitals/industries.
- **Vermicomposting:** Using earthworms (covered in detail below).

Conditions for Good Composting (CHON Formula)

- **C:N Ratio:** Ideal carbon-to-nitrogen ratio is 25–30:1. High carbon materials (browns) = dry leaves, wood chips; High nitrogen (greens) = food scraps, grass clippings.
- **Moisture:** 50–60% moisture content — like a wrung-out sponge.
- **Aeration:** Regular turning provides oxygen for aerobic decomposition and prevents odor.
- **Temperature:** 55–65°C during active phase kills pathogens and weed seeds.
- **Particle Size:** Smaller particles decompose faster (2–5 cm ideal).

Advantages of Composting

- Converts organic waste to valuable soil amendment (reduces landfill pressure).
- Improves soil structure, water-holding capacity, and microbial activity.
- Reduces the need for chemical fertilizers.
- Reduces greenhouse gas emissions compared to landfill disposal.

5.3 Vermiculture (Vermicomposting)

Vermiculture is the cultivation of earthworms, while vermicomposting is the process of using earthworms to convert organic waste into vermicompost — a nutrient-rich, biologically active soil amendment. The most commonly used species are *Eisenia fetida* (red wiggler) and *Lumbricus rubellus*.

Process of Vermicomposting

- **Step 1 — Bed Preparation:** Construct wooden/plastic beds (vermibed). Add a layer of dry straw/leaves as bedding.
- **Step 2 — Feed Addition:** Add pre-composted organic waste (food scraps, agricultural residue). Avoid meat, dairy, oily foods.
- **Step 3 — Worm Inoculation:** Introduce earthworms at a density of 1–2 kg worms per 1 m² of bed.
- **Step 4 — Maintenance:** Maintain 60–70% moisture, temperature 20–30°C, and regular feeding.
- **Step 5 — Harvest:** After 45–60 days, vermicompost is ready — dark, crumbly, odorless material.

Characteristics of Vermicompost

- Rich in nitrogen, phosphorus, potassium (NPK) and micronutrients.
- Contains plant growth hormones (auxins, cytokinins) produced by earthworms.
- Improves soil aeration, water retention, and microbial diversity.
- Suppresses plant diseases (contains beneficial bacteria and fungi).

5.4 Landfills (Sanitary Landfills)

A sanitary landfill is an engineered facility for the disposal of solid waste on land, designed to minimize environmental impacts through controlled placement, compaction, and cover of waste.

Components of a Sanitary Landfill

- **Bottom Liner System:** Multiple layers of compacted clay and high-density polyethylene (HDPE) geomembrane to prevent leachate seeping into groundwater.
- **Leachate Collection System:** Network of perforated pipes at the base to collect contaminated liquid (leachate) formed as water percolates through waste.
- **Daily Cover:** Each day's waste is covered with 15 cm of soil or alternative cover material to control odor, pests, and litter.
- **Gas Collection System:** Pipes extract landfill gas (55–65% methane, 35–45% CO₂) for energy generation.
- **Groundwater Monitoring Wells:** Detect any leachate contamination of surrounding groundwater.
- **Final Cap/Cover:** When full, covered with clay, topsoil, and vegetation to prevent infiltration.

Disadvantages of Landfills

- Landfill gas (methane) contributes to greenhouse effect if not captured.
- Leachate contamination remains a long-term risk even after closure.
- Requires large land areas — land becoming scarce near cities.
- Public opposition (NIMBY — Not In My Back Yard syndrome).

5.5 Hazardous Waste Treatment

Hazardous waste is waste that possesses properties making it dangerous or capable of having a harmful effect on human health or the environment. Properties include: toxicity, flammability, corrosivity, and reactivity.

Treatment Technologies for Hazardous Waste

- **High-Temperature Incineration:** Burning at 850–1200°C destroys organic hazardous compounds. Produces ash, flue gases (requires scrubbing). Used for chlorinated solvents, PCBs, pesticides.
- **Chemical Treatment:** Neutralization (acid + base), precipitation, oxidation/reduction to convert hazardous to less harmful forms.
- **Stabilization/Solidification:** Mixing waste with cement, lime, or polymer to immobilize contaminants, reducing leachability.
- **Secure Landfill (RCRA Cell):** Double-lined cells with extensive leachate monitoring for residuals that cannot be treated.
- **Plasma Arc Treatment:** Ultra-high temperature (5,000–15,000°C) plasma torch converts hazardous organic waste into syngas. Very effective but expensive.

PART F: BIOREMEDIATION TECHNOLOGIES & PHYTOREMEDIATION

6.1 Bioremediation

Bioremediation is the use of living organisms — primarily microorganisms (bacteria, fungi, algae) — to degrade, detoxify, transform, or immobilize environmental pollutants (organic and some inorganic) to less toxic or non-toxic forms.

Principles of Bioremediation

- **Biodegradation:** Microorganisms use pollutants as carbon/energy sources, converting them to CO₂, water, and biomass.
- **Bioaccumulation:** Organisms accumulate pollutants (e.g., heavy metals) internally.
- **Biotransformation:** Chemical transformation of pollutants to less toxic forms (not always complete mineralization).

Types of Bioremediation

In-situ Bioremediation: Treatment of contaminated soil/water at the site without excavation:

- **Bioventing:** Air injected into soil to stimulate aerobic biodegradation of petroleum hydrocarbons.
- **Biosparging:** Air/oxygen injected below water table to stimulate in-situ biodegradation.
- **Biostimulation:** Adding nutrients (nitrogen, phosphorus) or electron acceptors to stimulate native microbes.
- **Bioaugmentation:** Adding specialized microorganisms (cultured in lab) to contaminated site to enhance degradation.

Ex-situ Bioremediation: Excavating contaminated material for treatment elsewhere:

- **Land Farming:** Contaminated soil spread on surface, tilled periodically to stimulate aerobic degradation by native microorganisms.
- **Biopiles:** Contaminated soil piled and aerated; nutrients and moisture added.
- **Bioreactors:** Controlled vessel (slurry reactor) with contaminated material, microorganisms, nutrients — optimized conditions.

Factors Affecting Bioremediation

- **Temperature:** Optimal 20–40°C for most bacteria; low temperatures slow microbial activity.
- **pH:** Neutral pH (6–8) is optimal for most degrading bacteria.
- **Oxygen:** Aerobic degradation is faster; some contaminants require anaerobic conditions.
- **Nutrient Availability:** N, P required for microbial growth; often rate-limiting in contaminated sites.
- **Moisture:** 50–80% of field capacity optimal for soil bioremediation.
- **Contaminant Bioavailability:** Compounds bound to soil particles may not be available to microbes.

Key Microorganisms Used in Bioremediation

- **Pseudomonas putida:** Degrades petroleum hydrocarbons, toluene, naphthalene.
- **Alcanivorax borkumensis:** Marine bacteria; degrades alkanes; blooms after oil spills.
- **Dehalococcoides mccartyi:** Anaerobic; dechlorinates chlorinated solvents (TCE, PCE) to harmless ethene.
- **Geobacter metallireducens:** Reduces heavy metals (U, Cr, Fe) to less soluble forms.

- **Phanerochaete chrysosporium (white rot fungus):** Degrades lignin, PCBs, dioxins, PAHs.

Constructed Wetlands

Constructed wetlands are engineered systems designed to mimic natural wetland processes for treating wastewater, stormwater, and contaminated water. Plants, soils, and associated microorganisms work together to remove pollutants.

- **Surface Flow Wetlands:** Water flows above substrate surface; exposed to atmosphere; treats municipal wastewater.
- **Subsurface Flow Wetlands:** Water flows through gravel/sand bed below surface; less odor; better for cold climates.
- **Pollutant Removal Mechanisms:** Sedimentation (suspended solids), microbial degradation (BOD/COD), nitrification/denitrification (nitrogen removal), plant uptake (nutrients and some heavy metals).
- **Common Plants Used:** *Phragmites australis* (common reed), *Typha* spp. (cattails), *Scirpus* (bulrush).

6.2 Phytoremediation

Phytoremediation is the use of green plants and their associated microorganisms to degrade, contain, or remove pollutants from contaminated soil, water, or air. It is a cost-effective, solar-energy-driven, and aesthetically acceptable remediation approach.

Mechanisms of Phytoremediation

- **Phytoextraction (Phytoaccumulation):** Plants absorb pollutants (especially heavy metals) through roots and translocate to shoots/leaves. Biomass is harvested and disposed. Hyperaccumulators: *Thlaspi caerulescens* (zinc/cadmium), sunflower (uranium), Indian mustard (lead, selenium).
- **Phytostabilization:** Plants immobilize pollutants in soil or root zone, preventing spread. Heavy metal-tolerant grasses (e.g., vetiver) stabilize mine tailings.
- **Phytodegradation:** Plants and root-associated microbes metabolize/degrade organic pollutants (TCE, pesticides, explosives) within plant or rhizosphere.
- **Rhizofiltration:** Plant roots absorb, concentrate, and precipitate contaminants from water. Sunflowers used at Chernobyl to remove radiocesium from water.
- **Phytovolatilization:** Plants take up contaminants and release volatile forms through leaves. Mercury → methylmercury → volatile Hg^0 ; Selenium → volatile dimethylselenide.
- **Phytotransformation:** Plants enzymatically transform organic pollutants within plant tissues (e.g., TNT degradation by poplar trees).

Advantages of Phytoremediation

- Low cost compared to conventional technologies (3–8 times cheaper).
- Solar energy-driven — no external energy input required.
- Aesthetically pleasing, publicly acceptable, ecologically beneficial.
- Applicable to large contaminated areas.
- Preserves soil structure and microbial community.

Limitations of Phytoremediation

- Slow — takes months to years to achieve cleanup goals.

- Limited to shallow contamination (root depth of plants).
- Seasonal limitations in cold climates.
- Harvested biomass containing heavy metals requires specialized disposal.

PART G: DEGRADATION OF XENOBIOTICS IN ENVIRONMENT

7.1 What are Xenobiotics?

Xenobiotics (from Greek *xenos* = foreign, *bios* = life) are chemical substances foreign to biological systems — synthetic compounds not produced by nature or present in concentrations far exceeding their natural occurrence. Examples include pesticides, synthetic drugs, plastics, dyes, and industrial chemicals.

Why Xenobiotic Degradation Matters

- Xenobiotics persist in environment due to novel chemical structures that natural enzymes cannot easily break down.
- They bioaccumulate in food chains (biomagnification) reaching toxic concentrations in top predators.
- Understanding their degradation pathways is essential for bioremediation and environmental management.

7.2 Petroleum Hydrocarbons

Petroleum hydrocarbons (PHCs) are organic compounds composed mainly of carbon and hydrogen, derived from crude oil. They include alkanes, cycloalkanes, aromatic hydrocarbons (benzene, toluene, ethylbenzene, xylene — BTEX), and polycyclic aromatic hydrocarbons (PAHs).

Environmental Fate of Petroleum Hydrocarbons

- **Volatilization:** Light fractions evaporate (pentane, hexane).
- **Dissolution:** Water-soluble fractions (BTEX) dissolve and contaminate groundwater.
- **Adsorption:** Bind to soil organic matter and clay particles.
- **Biodegradation:** Microbial degradation is the primary natural attenuation mechanism.

Microbial Degradation of Petroleum Hydrocarbons

Hydrocarbon-degrading bacteria (e.g., *Pseudomonas*, *Arthrobacter*, *Acinetobacter*, *Alcanivorax*) use hydrocarbons as carbon and energy sources. Aerobic degradation proceeds via:

- **Alkane Degradation:** Terminal oxidation by monooxygenases → primary alcohol → aldehyde → fatty acid → beta-oxidation → $\text{CO}_2 + \text{H}_2\text{O}$.
- **Aromatic (BTEX) Degradation:** Dioxygenases introduce two oxygen atoms into the ring → dihydrodiol → catechol → ring cleavage → TCA cycle intermediates.
- **PAH Degradation:** Stepwise ring oxidation; higher molecular weight PAHs (pyrene, benzo[a]pyrene) are more recalcitrant.

7.3 Pesticides

Pesticides are chemical substances used to kill pests (insects, weeds, fungi, rodents). Major classes include organochlorines (DDT, BHC), organophosphates (malathion, chlorpyrifos), carbamates (carbaryl), and pyrethroids (cypermethrin).

Environmental Problems with Pesticides

- **Persistence:** Organochlorine pesticides (DDT) have half-lives of years to decades in soil and sediments.

- **Biomagnification:** DDT concentrates 10 million times from water (0.000003 ppm) to top predators like osprey (25 ppm). Causes reproductive failure (thinning of eggshells in birds).
- **Non-target effects:** Kill beneficial insects (bees, predatory insects), disrupting ecosystem services.
- **Groundwater contamination:** Leaching of soluble pesticides into aquifers.

Degradation of Pesticides

- **Microbial Degradation (Biotic):** Most important. Bacteria (e.g., *Flavobacterium* sp.) possess specific catabolic genes to degrade organophosphates (via phosphotriesterase enzymes). Fungi degrade complex structures via oxidative enzymes.
- **Photodegradation:** UV radiation breaks chemical bonds in pesticide molecules.
- **Hydrolysis:** Chemical reaction with water degrades ester linkages in organophosphates and carbamates.
- **Oxidation/Reduction:** Soil redox conditions affect pesticide transformation (e.g., DDT → DDD under anaerobic conditions).

7.4 Heavy Metals

Heavy metals are metallic elements with high atomic weight and density greater than 5 g/cm³. Environmentally important heavy metals include lead (Pb), mercury (Hg), cadmium (Cd), arsenic (As), chromium (Cr), and nickel (Ni).

Sources of Heavy Metal Pollution

- **Mining & Smelting:** Release of Pb, Zn, Cu, As from ore processing.
- **Industrial Effluents:** Electroplating (Cr, Ni, Cu), battery manufacturing (Pb, Cd).
- **Agriculture:** Phosphate fertilizers contain Cd; sewage sludge applied to fields contains multiple heavy metals.
- **Combustion:** Coal burning releases Hg, Pb; incineration releases Cd, Pb.
- **Paints & Pigments:** Lead-based paints, cadmium yellow pigment.

Health Effects of Key Heavy Metals

- **Lead (Pb):** Neurotoxin — cognitive impairment, behavioral problems, anemia in children. No safe level of exposure.
- **Mercury (Hg):** Minamata disease (Japan, 1956) — methylmercury causes severe neurological damage. Bioaccumulates strongly in fish.
- **Cadmium (Cd):** Itai-Itai disease (Japan) — bone softening, kidney failure. Carcinogenic.
- **Arsenic (As):** Skin lesions, peripheral neuropathy, bladder/lung/skin cancer. Chronic groundwater contamination issue in Bangladesh.

Transformation of Heavy Metals by Microorganisms

- **Biosorption:** Cell walls of bacteria/fungi bind heavy metals, removing them from solution.
- **Bioprecipitation:** Sulfate-reducing bacteria produce H₂S which precipitates metals as insoluble sulfides.
- **Biooxidation/Bioreduction:** *Geobacter* reduces soluble Cr(VI) to insoluble Cr(III); *Thiobacillus* oxidizes Fe²⁺ to Fe³⁺ (used in bioleaching).
- **Methylation:** Bacteria convert inorganic Hg to volatile methylmercury (more toxic but also mobilized).

PART H: E-WASTE — INTRODUCTION, TYPES, IMPACT, RECYCLING & RULES

8.1 Introduction to E-Waste

Electronic waste (e-waste) or Waste Electrical and Electronic Equipment (WEEE) refers to discarded electrical or electronic devices. It is the fastest-growing solid waste stream in the world, with approximately 53.6 million metric tons generated globally in 2019.

India is the third-largest e-waste generator in Asia (~3.2 million tonnes/year). Only 17–20% of global e-waste is formally recycled.

Why E-Waste is a Concern

- Contains hundreds of hazardous materials: heavy metals (Pb, Hg, Cd, Cr, Be), flame retardants (PBDEs), PVC.
- Also contains valuable and recoverable materials: gold, silver, palladium, copper, rare earth elements.
- Informal recycling (burning cables, acid baths) causes severe health and environmental damage.

8.2 Types of E-Waste

Category-wise Classification

- **Large Household Appliances:** Refrigerators, washing machines, air conditioners, microwaves.
- **Small Household Appliances:** Toasters, vacuum cleaners, electric shavers, hair dryers.
- **IT & Telecommunications Equipment:** Computers, laptops, mobile phones, printers, fax machines, photocopiers.
- **Consumer Electronics:** Televisions, DVD players, audio equipment, cameras.
- **Lighting Equipment:** Fluorescent lamps, LED bulbs, high-intensity lamps (contain Hg).
- **Electrical & Electronic Tools:** Drills, saws, sewing machines, welding equipment.
- **Toys, Leisure & Sports Equipment:** Electronic games, exercise equipment with electrical components.
- **Medical Devices:** Diagnostic equipment, radiotherapy devices, dialysis equipment.
- **Monitoring & Control Instruments:** Smoke detectors, thermostats, laboratory instruments.

8.3 Environmental Impact of E-Waste

Soil Contamination

- Lead from CRT (cathode ray tube) monitors leaches into soil; affects soil microbial communities.
- Cadmium from batteries and circuit boards — highly persistent, accumulates in food crops.
- Mercury from flat panel displays contaminates soil and converts to methylmercury.

Water Contamination

- Acid bath processes (used to extract gold informally) create acidic wastewater containing dissolved metals.
- Leaching from informal dump sites contaminates groundwater with Pb, Cd, Hg, As.
- Cyanide used in gold extraction poses extreme aquatic toxicity.

Air Pollution

- **Cable burning:** Releases dioxins, furans, and particulate matter (Pb, Cd) — highly toxic and carcinogenic.
- **Circuit board heating:** Releases brominated flame retardants (PBDEs) and polychlorinated biphenyls (PCBs).
- **Melting plastics:** Releases benzene, styrene, and other VOCs.

Health Impacts on Informal Workers

- Lead poisoning — impaired cognitive development in children living near e-waste sites.
- Mercury exposure — neurological damage, kidney failure.
- Chromium VI exposure — skin ulceration, carcinogenicity.
- Exposure to dioxins — endocrine disruption, cancer risk.
- Studies at Guiyu (China), Agbogbloshie (Ghana) — major informal e-waste sites — show extremely high heavy metal body burdens in workers and children.

8.4 E-Waste Recycling

E-waste recycling involves collecting, sorting, dismantling, and processing discarded electronics to recover valuable materials and safely dispose of hazardous ones.

Formal Recycling Process

- **Collection:** Drop-off centers, extended producer responsibility (EPR) programs, take-back schemes.
- **Sorting & Dismantling:** Manual disassembly to separate components (batteries, CRTs, circuit boards, cables, plastic housing).
- **Mechanical Processing:** Shredding, magnetic separation (ferrous metals), eddy current separation (non-ferrous metals), gravity separation.
- **Hydrometallurgy:** Acid/base leaching, solvent extraction, electrowinning to recover gold, silver, palladium from PCBs.
- **Pyrometallurgy:** High-temperature smelting to recover copper, precious metals from mixed e-waste.
- **Hazardous Component Treatment:** CRT glass (Pb) — smelted; Hg lamps — retorting to recover Hg; batteries — specialized recycling.

Urban Mining Potential

- One tonne of mobile phones contains 250g gold — compared to 5g per tonne of gold ore.
- Global e-waste contains ~57 billion USD worth of recoverable materials (2019 estimate).
- Recycling 1 million laptops saves energy equivalent to electricity used by 3,500 US homes for a year.

8.5 E-Waste Management Rules in India

E-Waste (Management) Rules, 2016 — Key Provisions

- **APPLIES TO:** Manufacturers, producers, dealers, refurbishers, consumers, and dismantlers/recyclers of EEE (Electrical & Electronic Equipment).
- **EXTENDED PRODUCER RESPONSIBILITY (EPR):** Producers responsible for financing/organizing e-waste collection and recycling. Must meet collection targets (10% of sales in first 2 years, increasing by 10% annually).
- **DEPOSIT REFUND SCHEME:** Deposits collected at sale; refunded when products returned for recycling.

- AUTHORIZED COLLECTION CENTERS: Dealers must set up collection centers or link with authorized dismantlers/recyclers.
- CHANNELIZATION: E-waste must be channelized to authorized recyclers — informal recycling is illegal.
- STATE POLLUTION CONTROL BOARD (SPCB): Authorizes dismantlers and recyclers; enforces compliance.
- REDUCTION OF HAZARDOUS SUBSTANCES (RoHS): Products must comply with limits on Pb, Hg, Cd, Cr-VI, PBB, PBDE.
- PENALTIES: Non-compliance may result in cancellation of authorization, fines under Environment Protection Act, 1986.
- AMENDMENT (2022): Strengthened EPR targets; introduced Producer Responsibility Organizations (PROs) to facilitate EPR compliance.

IMPORTANT DIFFERENCE TABLES

Composting vs. Vermicomposting

Composting	Vermicomposting
Decomposition by microorganisms (bacteria, fungi)	Decomposition by earthworms AND microorganisms
Process is purely microbial and thermophilic (55-65°C)	Mesophilic process (20-30°C); worms avoid high temperatures
Takes 2–6 months depending on method	Faster: 45–60 days
No special organisms required; uses natural microbial community	Requires introduction/maintenance of earthworm populations
Temperature kills pathogens and weed seeds naturally	May require pre-composting to destroy pathogens
Produces bulk soil amendment	Produces high-quality, nutrient-rich vermicompost with growth hormones
Can handle large volumes of waste	Limited by earthworm population and management needs

In-situ vs. Ex-situ Bioremediation

In-situ Bioremediation	Ex-situ Bioremediation
Treatment at the contamination site without excavation	Contaminated material excavated and treated at another location
Less expensive; no excavation/transportation costs	More expensive due to excavation, transport, and facility costs
Less disruptive to site; minimal disturbance	Highly disruptive; excavation creates secondary risks
Difficult to control conditions (temperature, pH, nutrients)	Conditions can be optimized (bioreactor control)
Slower treatment times generally	Generally faster treatment achievable
Examples: bioventing, biosparging, biostimulation	Examples: land farming, biopiles, slurry bioreactors
Suitable for deep contamination or sensitive sites	Suitable for highly contaminated small areas needing fast treatment

Phytoremediation vs. Bioremediation

Phytoremediation	Bioremediation
Uses plants (and associated microorganisms)	Uses microorganisms primarily (bacteria, fungi, algae)
Applicable to soil, water, and sediments	Applicable to soil, water, sediments, and subsurface

Cost effective; aesthetically pleasing	Can be cost-effective; less aesthetically visible
Limited to root zone depth (shallow contamination)	Can address deeper contamination (bioventing, biosparging)
Seasonal limitations; climate-dependent	Microbes function year-round (though slower in cold)
Excellent for heavy metals (phytoextraction)	Better suited for organic contaminant degradation
Requires harvesting and disposing of metal-laden biomass	Contaminants mineralized to CO ₂ /H ₂ O; no biomass issue

Noise vs. Sound

Sound	Noise
Vibrations that travel through a medium in organized patterns	Unwanted or unpleasant sound; irregular vibrations
Can be pleasant, meaningful, or neutral	Always unpleasant or disturbing to the receiver
Measured by frequency (Hz) and amplitude (dB)	Measured by loudness (dB), also considering context
Subjective judgment not implied	Judgment of unwantedness is key; same sound can be noise to one and not another
Used for communication, music, signals	Causes health and environmental problems

Formal vs. Informal E-Waste Recycling

Formal E-Waste Recycling	Informal E-Waste Recycling
Authorized recycling facilities with environmental controls	Unregulated backyard operations; illegal
Uses mechanical and hydrometallurgical processes	Uses acid stripping, open burning, manual dismantling
Workers protected with PPE; occupational health standards	Workers (often children) with no PPE; severe health risks
Hazardous materials properly handled and disposed	Toxic fumes, acid wastewater discharged to environment
Higher material recovery costs but safe and legal	Lower cost but creates massive environmental/health damage
Complies with e-waste management rules and EPR	Violates e-waste regulations; penalties apply

7-MARK QUESTIONS & MODEL ANSWERS

The following questions are typical 7-mark examination questions covering the entire Unit 3 syllabus. Each answer is structured for maximum marks.

Q1. Explain the causes, effects, and control measures of soil pollution. [7 Marks]

Definition: Soil pollution is the addition of harmful substances to the soil that degrade its quality and make it unfit for plant growth and human use.

Causes: (i) Industrial effluents and mining discharge heavy metals (Pb, Hg, Cd) into soil. (ii) Excessive use of synthetic fertilizers and pesticides reduces soil microbial diversity. (iii) Urban solid waste and plastic accumulation. (iv) Acid rain (H_2SO_4 , HNO_3) acidifies soil and leaches nutrients. (v) Radioactive waste contamination from nuclear facilities.

Effects: (i) Loss of soil fertility due to destruction of beneficial microorganisms. (ii) Bioaccumulation of heavy metals and pesticides in food chains. (iii) Groundwater contamination via leaching. (iv) Reduced crop yields and food safety concerns. (v) Desertification and loss of biodiversity. (vi) Human health impacts — cancer, neurological disorders, kidney damage.

Control Measures: (i) Integrated Pest Management (IPM) and organic farming to reduce chemical input. (ii) Proper solid waste management — composting, recycling, engineered landfills. (iii) Bioremediation using microorganisms to degrade pollutants. (iv) Phytoremediation using hyperaccumulator plants for heavy metal extraction. (v) Legislation: Environmental Protection Act, hazardous waste rules. (vi) Soil washing and solidification techniques for highly contaminated industrial sites.

Q2. Describe the sources, effects, and control of noise pollution with permissible limits. [7 Marks]

Definition: Noise pollution is unwanted sound that exceeds acceptable limits, causing harmful effects on humans, animals, and the environment.

Sources: (i) Transportation — road traffic, railways, aircraft are major urban noise sources. (ii) Industrial machinery — compressors, generators operating near residential areas. (iii) Construction activities — pile driving, drilling, demolition. (iv) Social events — loudspeakers, firecrackers during festivals. (v) Household appliances — AC, TV, mixers.

Effects on Humans: (i) Auditory: Tinnitus, Noise-Induced Hearing Loss (NIHL), deafness at sustained exposure above 85 dB. (ii) Cardiovascular: Elevated blood pressure, heart disease risk. (iii) Psychological: Stress, anxiety, sleep disturbance, irritability. (iv) Cognitive: Impaired learning and concentration in children.

Effects on Wildlife: Disrupts communication, feeding, and mating in animals; causes whale/dolphin beaching.

Permissible Limits (India, Noise Rules 2000): Industrial area: 75 dB (day)/70 dB (night); Commercial: 65/55 dB; Residential: 55/45 dB; Silence zone: 50/40 dB.

Control: (i) Source control — silencers, anti-vibration mounts. (ii) Path control — sound barriers, green belts, soundproofing. (iii) Receiver control — ear protection devices. (iv) Zoning — industrial areas away from residences. (v) No-honking zones near hospitals and schools. (vi) Public awareness campaigns.

Q3. What are nuclear and radiation hazards? Discuss their sources, health effects, and control measures. [7 Marks]

Definition: Nuclear and radiation hazards refer to the risks associated with ionizing radiation (alpha, beta, gamma, neutron) and radioactive materials that damage living tissues and contaminate the environment.

Sources: (i) Nuclear power plants — routine emissions and major accidents (Chernobyl 1986: 600,000 people affected; Fukushima 2011). (ii) Nuclear weapons testing — atmospheric tests spread radioactive fallout (Sr-90, Cs-137) globally. (iii) Medical applications — X-rays, CT scans, radiotherapy, nuclear medicine. (iv) Industrial uses — radioactive gauges, sterilization equipment. (v) Mining of uranium/thorium — releases radon gas. (vi) Natural background — cosmic rays, radon from granite rocks.

Somatic Health Effects: (i) Acute Radiation Syndrome (ARS) — doses >1 Gy cause nausea, bone marrow failure, death. (ii) Cancer — leukemia, thyroid cancer, lung cancer are most common. (iii) Cataracts of eye lens. (iv) Radiation burns and skin ulceration.

Genetic Effects: DNA mutations and chromosomal aberrations in germ cells transmitted to future generations causing birth defects and hereditary diseases.

Environmental Effects: Long-term contamination of large areas; bioaccumulation of Sr-90 in bones and Cs-137 in muscles through food chains.

Control: (i) Three principles of radiation protection: Time (minimize), Distance (maximize), Shielding (use lead/concrete). (ii) Personal dosimeters to monitor exposure. (iii) Safe waste disposal — near-surface for low-level, deep geological repositories for high-level waste. (iv) International treaties — NPT, CTBT. (v) Emergency preparedness and evacuation plans. (vi) Iodine tablet distribution to prevent thyroid radiation absorption.

Q4. Explain ocean acidification — causes, chemistry, effects, and solutions. [7 Marks]

Definition: Ocean acidification is the progressive decrease in ocean pH due to absorption of excess atmospheric CO₂ produced by human activities.

Cause: Oceans absorb approximately 30% of anthropogenic CO₂ emissions (~10 billion tonnes/year). Since industrialization, ocean pH has fallen from 8.2 to ~8.1 — a 26% increase in acidity on the logarithmic pH scale.

Chemistry: $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$ (carbonic acid) $\rightarrow \text{H}^+ + \text{HCO}_3^-$ (bicarbonate). The increase in H⁺ ions reduces carbonate ion (CO₃²⁻) concentration, which is essential for calcification by marine organisms.

Effects: (i) Calcification Impairment: Oysters, mussels, sea urchins, and pteropods cannot build/maintain calcium carbonate shells — shells dissolve. (ii) Coral Reef Destruction: Corals cannot build skeletons; projected loss of 70% of cold-water corals if CO₂ continues rising. (iii) Food Web Disruption: Loss of pteropods (key food source for salmon, herring, whales) cascades through food chains. (iv) Fish Behavior: Impairs olfactory senses used to detect predators; affects otolith (ear stone) development. (v) Phytoplankton: Some calcifying species (coccolithophores) less competitive in acidified waters.

Solutions: (i) Reduce CO₂ emissions — primary solution: renewable energy, energy efficiency, carbon capture. (ii) Ocean alkalinity enhancement (experimental). (iii) Marine protected areas to reduce additional stressors. (iv) Seaweed aquaculture to locally buffer acidity. (v) International cooperation under Paris Agreement and UNFCCC.

Q5. Describe bioremediation technologies used for pollution control. Include types and key microorganisms. [7 Marks]

Definition: Bioremediation is the use of living organisms (bacteria, fungi, plants) to degrade, detoxify, or immobilize pollutants in contaminated environments.

In-situ Technologies (Treatment at site): (i) Bioventing: Air injected into unsaturated soil to stimulate aerobic degradation of petroleum hydrocarbons. (ii) Biosparging: Air/O₂ injected below water table to enhance in-situ aerobic degradation. (iii) Biostimulation: Adding nutrients (N, P) to stimulate indigenous microorganisms. (iv) Bioaugmentation: Introducing laboratory-cultured specialized microorganisms.

Ex-situ Technologies (Excavated material treated elsewhere): (i) Land Farming: Contaminated soil spread on lined area, tilled for aeration; relies on native microbes. (ii) Biopiles: Contaminated soil in engineered piles with aeration pipes. (iii) Slurry-phase Bioreactors: Soil mixed with water in controlled reactor — fastest method.

Constructed Wetlands: Engineered systems using plants (Phragmites, Typha) and associated microorganisms to remove BOD, nitrogen, and some metals from wastewater.

Key Microorganisms: (i) *Pseudomonas putida* — petroleum hydrocarbons, toluene. (ii) *Alcanivorax borkumensis* — marine oil spill response. (iii) *Dehalococcoides mccartyi* — dechlorinates chlorinated solvents (TCE) anaerobically. (iv) *Geobacter metallireducens* — reduces heavy metals (uranium, chromium). (v) *Phanerochaete chrysosporium* (white rot fungus) — degrades PAHs, dioxins, PCBs via lignin-peroxidase system.

Factors Affecting Bioremediation: Temperature, pH, oxygen availability, nutrient levels, moisture content, and contaminant bioavailability.

Q6. What is phytoremediation? Explain its mechanisms with suitable examples. [7 Marks]

Definition: Phytoremediation uses green plants and associated root-zone microorganisms to remove, degrade, or immobilize pollutants from contaminated soil, water, or air.

Phytoextraction: Plants absorb heavy metals through roots and translocate to harvestable shoots. Examples: *Thlaspi caerulescens* (zinc, cadmium hyperaccumulator — up to 3% Zn in dry weight), sunflower (uranium extraction from water near Chernobyl), *Brassica juncea* (lead, selenium).

Phytostabilization: Plants reduce pollutant mobility in soil without accumulation in shoots. Metal-tolerant grasses (vetiver, *Agrostis*) planted on mine tailings to prevent wind/water dispersal.

Phytodegradation: Plants and rhizosphere microorganisms enzymatically degrade organic pollutants. Poplar trees degrade TNT (explosive) and TCE (solvent); willow trees treat atrazine (herbicide).

Rhizofiltration: Plant roots absorb/adsorb contaminants from water. Sunflowers used at Chernobyl to remove radiocesium and radiostrontium from water bodies.

Phytovolatilization: Plants take up and release volatile forms through leaves. Transgenic plants expressing mercuric reductase convert methylmercury to less toxic volatile Hg^0 .

Advantages: Cost-effective (3-8x cheaper than conventional methods), solar-powered, aesthetically pleasing, preserves soil ecosystem. **Limitations:** Slow (years), shallow depth, seasonal, harvested biomass requires disposal.

Q7. Explain e-waste in detail — types, environmental impact, recycling process, and management rules in India. [7 Marks]

Definition: E-waste (Waste Electrical and Electronic Equipment, WEEE) refers to discarded electrical/electronic devices. Global generation reached 53.6 million metric tonnes in 2019; India generates ~3.2 million tonnes annually.

Types: Large appliances (refrigerators, AC); Small appliances (toasters, shavers); IT/Telecom equipment (computers, mobiles, printers); Consumer electronics (TVs, audio); Lighting (fluorescent lamps with Hg); Medical devices; Monitoring instruments.

Environmental Impact: (i) Soil: Pb from CRTs, Cd from batteries, Hg from displays leach into soil and persist for decades. (ii) Water: Acid bath leachate and landfill leachate contaminate groundwater with multiple heavy metals. (iii) Air: Cable burning releases dioxins, furans, Pb particulates; plastic melting releases benzene, styrene.

Health Impact: Informal recycling workers (often children) suffer lead poisoning (cognitive impairment), mercury toxicity (neurological damage), chromium VI exposure (carcinogenic), and dioxin-related endocrine disruption.

Formal Recycling Process: Collection → Sorting and dismantling → Mechanical processing (shredding, magnetic/eddy current separation) → Hydrometallurgy (acid leaching, solvent extraction, electrowinning for gold/silver/palladium) → Hazardous component treatment (CRT glass smelting, Hg retorting).

E-Waste (Management) Rules, 2016 — India: Extended Producer Responsibility (EPR) — producers must finance collection/recycling meeting mandatory targets; Deposit Refund Scheme; Only authorized recyclers permitted; RoHS compliance (limits on Pb, Hg, Cd, Cr-VI, PBBs, PBDEs); Dealers establish collection centers; SPCB authorizes dismantlers. 2022 Amendment strengthened EPR targets and introduced Producer Responsibility Organizations (PROs).

Q8. Write a comprehensive note on Oil Spills — sources, effects on marine ecosystem, and cleanup methods. [7 Marks]

Definition: An oil spill is the unintended release of liquid petroleum into the ocean or coastal waters, constituting one of the most damaging forms of marine pollution.

Sources: (i) Tanker accidents — Exxon Valdez (1989, 37,000 tonnes, Prince William Sound, Alaska); Deepwater Horizon (2010, largest marine oil spill, 4.9 million barrels). (ii) Offshore platform blowouts. (iii) Routine shipping operations — bilge pumping, tank washing. (iv) Pipeline corrosion and leaks. (v) Natural submarine seeps (~600,000 tonnes/year globally).

Effects on Marine Ecosystem: (i) Seabirds: Oil destroys waterproofing of feathers causing hypothermia and drowning; ingested oil causes internal organ damage. (ii) Marine mammals: Sea otters and seals lose fur insulation; whales inhale hydrocarbon vapors damaging lungs. (iii) Fish: PAHs kill eggs and larvae; oxygen depletion in slick zone suffocates organisms. (iv) Coral reefs: Smothering and bleaching of coral polyps. (v) Mangroves: Oil clogs pneumatophore (breathing root) pores, killing trees. (vi) Fisheries: Massive economic losses; contaminated seafood.

Cleanup Methods: (i) Containment booms — float on surface to contain slick; especially effective in calm conditions. (ii) Skimmers — mechanical surface collection (weir, disc, drum type). (iii) Sorbents — natural (straw, peat) and synthetic materials absorb oil; recovery by compression. (iv) Chemical dispersants — break oil into micro-droplets (Corexit — controversial toxicity concerns). (v) In-situ burning — controlled burning in early stages; effective but creates air pollution. (vi) Bioremediation — nutrient addition (biostimulation) stimulates *Alcanivorax*, *Marinobacter*; bioaugmentation with oil-degrading bacteria. (vii) Shoreline cleanup — manual/mechanical removal of oiled sediment.

Q9. Describe the degradation of xenobiotics in the environment with special reference to petroleum hydrocarbons and pesticides. [7 Marks]

Xenobiotics: Synthetic chemical compounds foreign to biological systems that resist natural degradation due to their novel molecular structures. Key examples include petroleum hydrocarbons, chlorinated pesticides, PCBs, and plastics.

Petroleum Hydrocarbon Degradation: Aerobic biodegradation is the primary mechanism. (i) Alkane degradation: Terminal oxidation by monooxygenases produces primary alcohols → aldehydes → fatty acids → beta-oxidation to CO₂ and H₂O. Key organisms: *Alcanivorax*, *Pseudomonas*, *Rhodococcus*. (ii) Aromatic (BTEX) degradation: Dioxygenases attack the aromatic ring producing dihydrodiols → catechol → ring cleavage → TCA cycle intermediates. (iii) PAH degradation: Sequential ring oxidation; higher MW PAHs (benzo[a]pyrene) are highly recalcitrant. Factors affecting rate: Temperature, O₂ availability, nutrient levels (N, P), bioavailability (sorption to soil).

Pesticide Degradation: (i) Microbial degradation: *Flavobacterium* sp. produces phosphotriesterase enzyme to degrade organophosphates. *Pseudomonas* sp. dehalogenates organochlorines. *Phanerochaete* degrades complex aromatic pesticides via peroxidases. (ii) Photodegradation: UV light breaks chemical bonds in pesticide molecules at soil surface. (iii) Hydrolysis: Water molecules cleave ester bonds in organophosphates and carbamates — abiotic process. (iv) Oxidation/Reduction: Soil redox potential determines pathway; DDT reduced to DDD under anaerobic conditions.

Problem of Persistence: Organochlorine pesticides (DDT, HCH, dieldrin) have environmental half-lives of years because: (i) Highly hydrophobic — partition strongly into soil organic matter. (ii) No natural selection pressure for enzyme systems to degrade them. (iii) Biomagnification in food chains — 10-million-fold concentration from water to top predators. This persistence led to global bans under Stockholm Convention on Persistent Organic Pollutants (POPs), 2001.

QUICK REVISION SUMMARY

Unit 3 — Key Points to Remember

- **SOIL POLLUTION:** Causes = heavy metals, pesticides, acid rain. Control = bioremediation, phytoremediation, legislation.
- **SOIL WATER:** Types = gravitational, capillary (plant available), hygroscopic. Field capacity = max water after drainage.
- **NOISE POLLUTION:** Permissible limits = Residential 55/45 dB (day/night); Silence zones 50/40 dB. Units = decibels (dB).
- **NUCLEAR HAZARDS:** 3 principles of protection = Time, Distance, Shielding. High-level waste = deep geological repository.
- **OIL SPILLS:** Cleanup = booms → skimmers → dispersants → bioremediation (Alcanivorax). Deepwater Horizon = largest marine spill.
- **OCEAN ACIDIFICATION:** $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{H}^+ + \text{HCO}_3^-$. pH dropped from 8.2 to 8.1 = 26% more acidic. Destroys coral reefs.
- **COMPOSTING:** C:N ratio = 25-30:1; Temperature = 55-65°C. **VERMICOMPOSTING:** Worms + microbes, 45-60 days, mesophilic (20-30°C).
- **BIOREMEDIATION:** In-situ (bioventing, biosparging) vs Ex-situ (land farming, bioreactors). Key bugs: Pseudomonas, Alcanivorax, Dehalococcoides.
- **PHYTOREMEDIATION:** Phytoextraction (hyperaccumulators absorb metals), Phytostabilization, Phytodegradation, Rhizofiltration, Phytovolatilization.
- **E-WASTE RULES 2016:** EPR (producers responsible), Deposit refund scheme, RoHS compliance, Only authorized recycling permitted, SPCB oversight.
- **HEAVY METALS:** Pb = neurological damage; Hg = Minamata disease; Cd = Itai-Itai disease; As = skin lesions/cancer.
- **XENOBIOTICS:** Petroleum degraded by Pseudomonas (aerobic); PAH = polycyclic aromatic hydrocarbons, most recalcitrant. DDT persists decades.

❖ All the Best for Your Examinations! ❖